Time vs Ensemble Average of Brownian Motion

A B.Tech Project Report Submitted in Partial Fulfillment of the Requirements for the Degree of

Bachelor of Technology in Engineering Physics

by

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Declaration

I hereby declare that the project report entitled **"Time vs Study of Ensemble Average in Brownian Motion"** submitted to the **Department of Physics, Indian Institute of Technology Guwahati**, in partial fulfillment of the requirements for the degree of **Bachelor of Technology in Engineering Physics**, is a record of original work carried out by me under the supervision of **Prof. Samudrajit Thapa**.

I further declare that the work reported in this project has not been submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institution or university.

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Abstract

This project presents a comprehensive numerical study of Brownian motion and its statistical properties using a discrete Langevin dynamics framework. A large ensemble of stochastic trajectories was simulated in one dimension, and multiple complementary statistical analyses were performed to characterize the diffusive behavior of the system.

The time-averaged mean squared displacement (TAMSD) was computed for individual trajectories and ensemble-averaged to verify the linear scaling law $\langle \delta^2(\Delta) \rangle = 2D\Delta$, consistent with normal diffusion. A scaling analysis yielded an anomalous exponent $\alpha \approx 1$, and the estimated diffusion coefficient $D \approx 0.5214$ was found to be in excellent agreement with the input value $D = 0.5$. The ergodicity breaking parameter confirmed that the system approaches ergodic behavior in the limit of small Δ/T.

The probability distribution of particle positions was verified to follow a Gaussian form at all times, with variance growing linearly as $\sigma^2(t) = 2Dt$. Displacement distributions at multiple lag times further confirmed the Gaussian and ergodic nature of the process. Together, these analyses establish that the simulated system faithfully reproduces the fundamental statistical properties of classical Brownian motion.

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List of Figures

1 Subset of 20 Brownian motion trajectories out of the total ensemble of 5000, showing position x as a function of time step. 11
2 TAMSD computed for all simulated trajectories (colored curves) along with the ensemble-averaged TAMSD (black curve) on a log–log scale. . 13
3 Comparison of the Ensemble-Averaged MSD (EAMSD, blue solid line) and the Ensemble-Averaged Time-Averaged MSD (EATAMSD, black dashed line) on a log–log scale. 15
4 Variation of the ergodicity breaking parameter $EB(\Delta)$ as a function of lag time Δ, along with the theoretical prediction $EB(\Delta) = 4\Delta/3T$. . . 19
5 Histogram of particle positions at final time $t = T$ fitted with a Gaussian distribution, along with the theoretical prediction $P(x, t)$. 21
6 Time evolution of variance $\sigma^2(t)$ obtained from simulation (colored curve) compared with the theoretical prediction $2Dt$ (dashed line). . . . 23
7 Probability distributions of particle positions at different time instances, along with the corresponding fitted Gaussian and theoretical curves $P(x, t)$. 25
8 Displacement distributions for different lag values along with fitted Gaussian and theoretical curves. 27
9 Combined plot of displacement distributions for all lag values, showing the progressive broadening with increasing lag time Δ. 27

Contents

1 Introduction **7**

2 Diffusive Characteristics and Physical Observables (with derivation) **8**
2.1 Derivation of Time-Averaged Mean Squared Displacement (TAMSD) . 8
2.2 Discrete Form (Used in Computation) 8
2.3 Meaning of Terms 9

3 Numerical Simulation of Brownian Motion **10**
3.1 Model and Theoretical Background 10
3.2 Simulation Details 10

4 Time-Averaged Mean Squared Displacement Analysis **12**
4.1 Theoretical Framework 12
4.2 Connection with Brownian Motion 12
4.3 Physical Interpretation 13

5 Comparison of Ensemble-Averaged MSD and TAMSD **14**
5.1 Theoretical Background 14
5.2 Expected Behavior 14
5.3 Physical Interpretation 15

6 Scaling Analysis and Estimation of Diffusion Coefficient **16**
6.1 Theoretical Background 16
6.2 Methodology 16
6.3 Results 16

7 Ergodicity and Ergodicity Breaking Analysis **18**
7.1 Concept of Ergodicity 18
7.2 Definition of Ergodicity Breaking Parameter 18
7.3 Theoretical Expectation 18
7.4 Results and Discussion 18
7.5 Physical Interpretation 19

8 Distribution of Particle Positions and Gaussian Analysis **20**
8.1 Theoretical Background 20
8.2 Methodology 20
8.3 Results 20
8.4 Physical Interpretation 21

9 Time Evolution of Variance and Diffusion Validation **22**
9.1 Theoretical Background 22
9.2 Methodology 22
9.3 Results 22
9.4 Physical Interpretation 22

10 Time Evolution of Probability Distribution Function **24**
10.1 Theoretical Background 24
10.2 Methodology 24
10.3 Results 24
10.4 Physical Interpretation 25

11 Displacement Distribution Analysis at Different Lag Times **26**
11.1 Theoretical Background 26
11.2 Methodology 26
11.3 Results and Discussion 28
11.4 Physical Interpretation 28

12 Results and Summary **29**

13 Conclusion and Future Scope **30**
13.1 Future Scope 30

1 Introduction

The motion of particles in a stochastic environment is a fundamental problem in statistical physics, with applications ranging from simple Brownian motion to complex intracellular transport. In many physical and biological systems, the dynamics of particles are governed by random forces, leading to trajectories that appear irregular but exhibit well-defined statistical properties when analyzed collectively.

One of the key approaches to understanding such motion is through the study of **particle displacements and their statistical distributions**. For an ideal diffusive process, the displacement of particles over a fixed time interval is expected to follow a Gaussian distribution, as predicted by classical diffusion theory. This provides a direct way to characterize the system through parameters such as the mean displacement and variance, which are related to drift and diffusion respectively.

In this project, a stochastic trajectory consisting of discrete position data is analyzed to investigate its statistical behavior. From the given trajectory points, displacements are constructed using a fixed step size, and an ensemble of these displacements is generated. The resulting distribution is then examined to determine whether it follows a Gaussian form.

By fitting the displacement distribution to a Gaussian function, key parameters such as the mean (μ) and standard deviation (σ) are extracted. These parameters provide insight into the underlying dynamics of the system: the mean indicates any systematic drift, while the variance reflects the strength of random fluctuations and is directly related to the diffusion coefficient.

Such analysis is widely used in modern research on stochastic processes and single-particle tracking, where deviations from Gaussian behavior can reveal more complex dynamics such as confinement, heterogeneity, or non-equilibrium effects. Therefore, this study not only verifies fundamental diffusion behavior but also serves as a basis for understanding more advanced stochastic systems [1, 2, 3].

2 Diffusive Characteristics and Physical Observables (with derivation)

For a stochastic process in one dimension, the position of a particle is represented as $x(t)$. The displacement over a time interval Δ is defined as

$$\Delta x = x(t + \Delta) - x(t). \tag{1}$$

To characterize the motion statistically, we consider the mean squared displacement (MSD),

$$\langle [x(t) - x(0)]^2 \rangle \sim t^\alpha, \tag{2}$$

where α is the scaling exponent and K_α is the generalized diffusion coefficient, such that

$$\langle [x(t) - x(0)]^2 \rangle = 2K_\alpha t^\alpha. \tag{3}$$

2.1 Derivation of Time-Averaged Mean Squared Displacement (TAMSD)

Instead of ensemble averaging over many trajectories, we perform time averaging over a single trajectory of length T.

Start with displacement at time t:

$$\Delta x(t, \Delta) = x(t + \Delta) - x(t). \tag{4}$$

Square the displacement:

$$[\Delta x(t, \Delta)]^2 = [x(t + \Delta) - x(t)]^2. \tag{5}$$

To average over the entire trajectory, integrate over all possible starting times $t \in [0, T - \Delta]$:

$$\delta^2(\Delta) = \frac{1}{T - \Delta} \int_0^{T-\Delta} [x(t + \Delta) - x(t)]^2 \, dt. \tag{6}$$

2.2 Discrete Form (Used in Computation)

For a trajectory with discrete points:

$$x_1, x_2, x_3, \ldots, x_N, \tag{7}$$

time becomes discrete: $t \to k$, and the integral becomes a summation.

Thus,

$$\delta_i^2(\Delta) = \frac{1}{N - \Delta} \sum_{k=1}^{N-\Delta} \left[x_i(k + \Delta) - x_i(k) \right]^2. \tag{8}$$

2.3 Meaning of Terms

- $\delta_i^2(\Delta) \rightarrow$ time-averaged mean squared displacement for trajectory i
- $N \rightarrow$ total number of data points
- $\Delta \rightarrow$ lag time (in steps)
- summation $\rightarrow$ averaging over all possible time intervals

3 Numerical Simulation of Brownian Motion

3.1 Model and Theoretical Background

The motion of a Brownian particle in one dimension can be described using the stochastic Langevin framework. In the absence of external forces and damping effects, the position of the particle evolves due to random thermal fluctuations.

In discrete form, the position update equation is given by

$$x_{i+1} = x_i + \Delta x_i \tag{9}$$

where the displacement Δx_i is modeled as a random variable drawn from a Gaussian distribution. For a diffusive process, the increment satisfies

$$\Delta x_i \sim \mathcal{N}(0,\ 2D\,dt) \tag{10}$$

Thus, the update equation becomes

$$x_{i+1} = x_i + \sqrt{2D\,dt}\ \eta_i \tag{11}$$

where

- D is the diffusion coefficient
- dt is the time step
- $\eta_i \sim \mathcal{N}(0, 1)$ is a standard normal random variable

This formulation ensures that the variance of displacement grows linearly with time, consistent with Brownian motion:

$$\langle x^2(t) \rangle = 2Dt \tag{12}$$

3.2 Simulation Details

A large number of independent trajectories were generated numerically using the above stochastic update rule. The parameters used in the simulation are:

- Number of trajectories: $N = 5000$
- Number of time steps: 1000
- Time step: $dt = 10^{-4}$
- Diffusion coefficient: $D = 0.5$

Each trajectory starts from the origin and evolves through successive random displacements.

The figure shows a subset of 20 trajectories out of the total ensemble.

Key observations:

- All trajectories originate from the same initial position $x = 0$, indicating identical initial conditions.

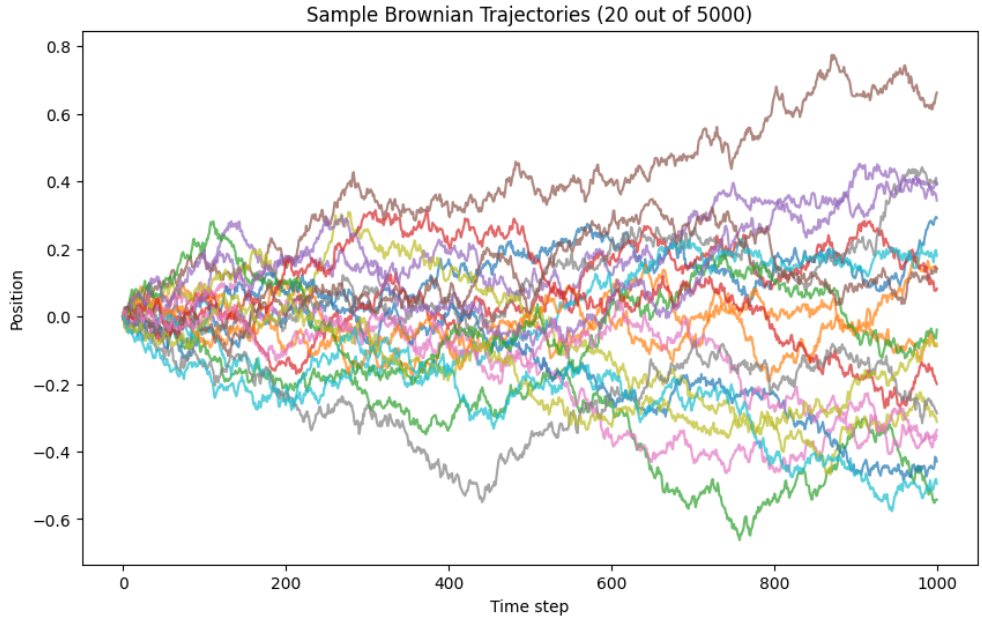

Figure 1: Subset of 20 Brownian motion trajectories out of the total ensemble of 5000, showing position x as a function of time step.

- As time progresses, trajectories spread out symmetrically in both positive and negative directions.
- The spread of trajectories increases with time, which is a direct signature of diffusive behavior.
- No preferred direction is observed, confirming that the mean displacement remains zero.
- The randomness in individual paths highlights the stochastic nature of motion, while the overall spread reflects underlying statistical regularity.

The increasing dispersion of trajectories with time is consistent with the theoretical prediction:

$$\langle x^2(t) \rangle \propto t \tag{13}$$

Thus, the simulation successfully reproduces the key features of Brownian motion, namely random trajectories with zero mean and variance growing linearly in time.

4 Time-Averaged Mean Squared Displacement Analysis

4.1 Theoretical Framework

To further quantify the stochastic nature of the simulated trajectories, we compute the **time-averaged mean squared displacement (TAMSD)**. Unlike the ensemble-averaged MSD, TAMSD is calculated along individual trajectories and provides insight into the time evolution of particle motion.

For a one-dimensional trajectory $x_i(t)$, the TAMSD is defined as

$$\delta_i^2(\Delta) = \frac{1}{N-\Delta} \sum_{k=1}^{N-\Delta} \left[x_i(k+\Delta) - x_i(k)\right]^2 \tag{14}$$

where

- Δ is the lag time,
- N is the total number of time steps,
- i labels the trajectory.

This expression represents the average of squared displacements over all possible time intervals of duration Δ within a single trajectory.

To obtain statistically robust results, we further compute the **ensemble-averaged TAMSD** over all trajectories:

$$\langle \delta^2(\Delta) \rangle = \frac{1}{N_{\text{traj}}} \sum_{i=1}^{N_{\text{traj}}} \delta_i^2(\Delta) \tag{15}$$

4.2 Connection with Brownian Motion

For an ideal Brownian process, the TAMSD follows the same scaling behavior as the MSD:

$$\langle \delta^2(\Delta) \rangle = 2D\Delta \tag{16}$$

Thus, on a log–log scale, the TAMSD is expected to exhibit a linear behavior with slope equal to 1, confirming normal diffusion.

The figure shows the TAMSD computed for all simulated trajectories along with the ensemble-averaged TAMSD (black curve).

Key observations:

- Individual trajectories (colored curves) exhibit fluctuations, especially at larger lag times, due to finite sampling and reduced statistics.
- At small lag times, all trajectories collapse onto a well-defined trend, indicating consistent diffusive behavior.
- The ensemble-averaged TAMSD (black line) shows a clear linear trend on the log–log scale.

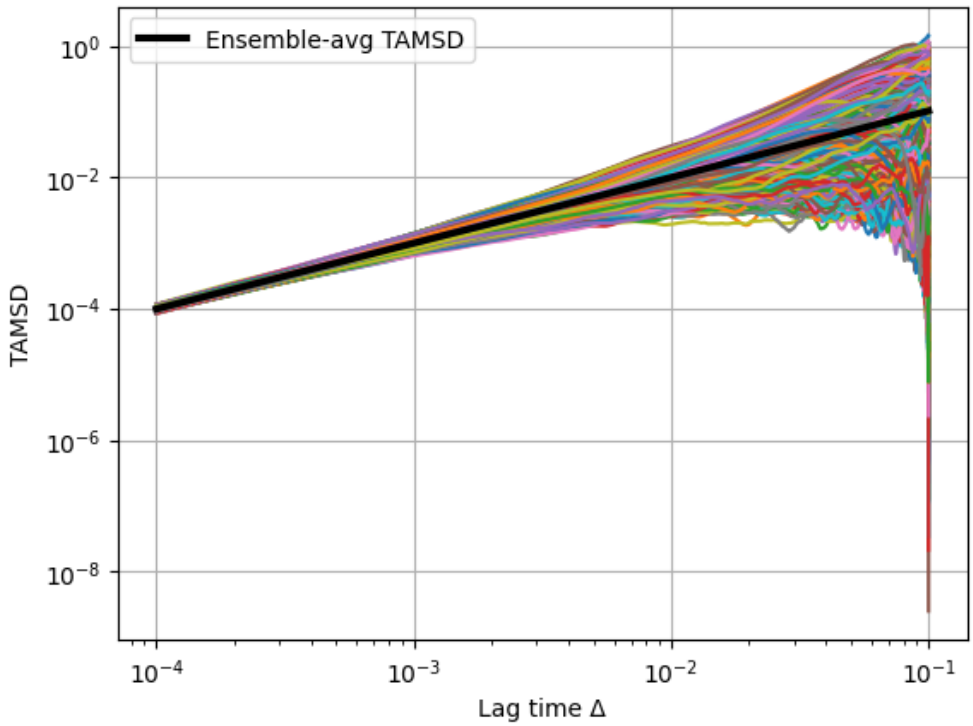

Figure 2: TAMSD computed for all simulated trajectories (colored curves) along with the ensemble-averaged TAMSD (black curve) on a log–log scale.

- This linear scaling confirms that

$$\delta^2(\Delta) \propto \Delta \tag{17}$$

  which is the hallmark of normal Brownian diffusion.

- Deviations at larger lag times arise because the number of available data points $(N - \Delta)$ decreases, leading to increased statistical noise.

4.3 Physical Interpretation

The agreement between individual TAMSDs and the ensemble-averaged behavior indicates that the system is **ergodic**, meaning time averages and ensemble averages are equivalent. This is a key property of standard Brownian motion.

Thus, the TAMSD analysis validates the numerical simulation and confirms that the generated trajectories correctly reproduce the fundamental characteristics of diffusive motion.

5 Comparison of Ensemble-Averaged MSD and TAMSD

5.1 Theoretical Background

To fully characterize the stochastic dynamics, we compare two fundamental quantities:

1. Ensemble-Averaged Mean Squared Displacement (EAMSD)

The ensemble-averaged MSD is defined as

$$\langle x^2(t)\rangle = \frac{1}{N_{\text{traj}}} \sum_{i=1}^{N_{\text{traj}}} [x_i(t) - x_i(0)]^2 \tag{18}$$

This represents averaging over all trajectories at a fixed time t.

2. Ensemble-Averaged Time-Averaged MSD (EATAMSD)

From the previous section, the time-averaged MSD for each trajectory is

$$\delta_i^2(\Delta) = \frac{1}{N-\Delta} \sum_{k=1}^{N-\Delta} [x_i(k+\Delta) - x_i(k)]^2 \tag{19}$$

Averaging this over all trajectories gives

$$\langle \delta^2(\Delta)\rangle = \frac{1}{N_{\text{traj}}} \sum_{i=1}^{N_{\text{traj}}} \delta_i^2(\Delta) \tag{20}$$

5.2 Expected Behavior

For standard Brownian motion, both quantities are expected to follow

$$\langle x^2(t)\rangle = 2Dt, \qquad \langle \delta^2(\Delta)\rangle = 2D\Delta \tag{21}$$

and importantly,

$$\langle x^2(t)\rangle = \langle \delta^2(t)\rangle \tag{22}$$

This equality indicates **ergodicity**, meaning time averages and ensemble averages are equivalent.

The figure shows the comparison between:

- **EAMSD (blue solid line)**
- **EATAMSD (black dashed line)**

Key observations:

- Both curves overlap almost perfectly over the entire time range.
- The linear behavior on the log–log plot confirms that

$$\text{MSD} \propto t \tag{23}$$

- The slope of the curves is approximately 1, consistent with normal diffusion.
- No deviation between EAMSD and EATAMSD is observed, even at larger times.

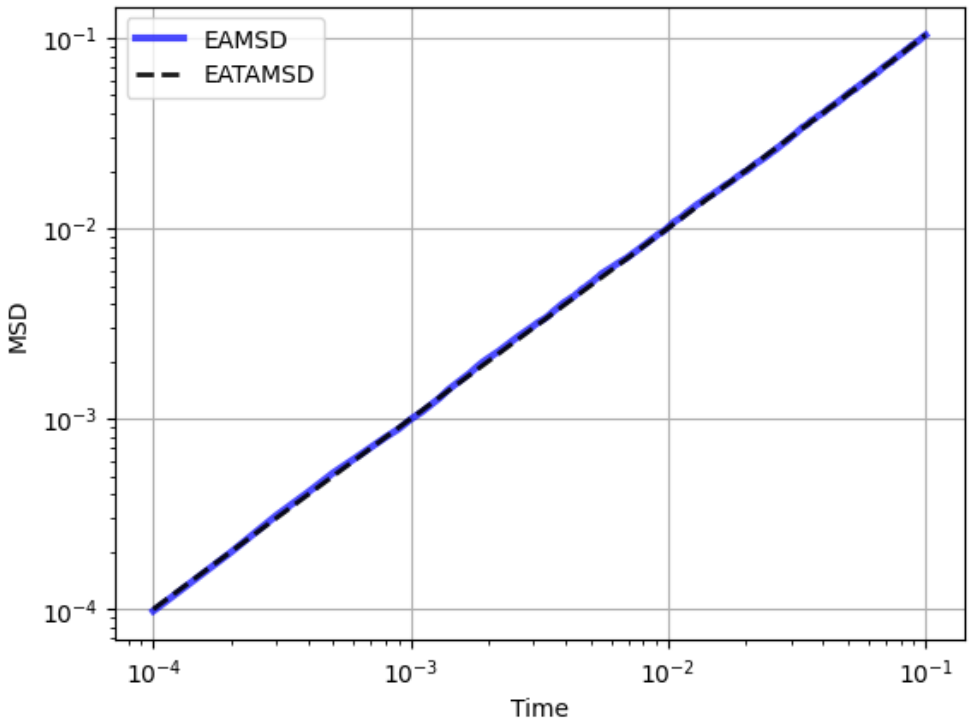

Figure 3: Comparison of the Ensemble-Averaged MSD (EAMSD, blue solid line) and the Ensemble-Averaged Time-Averaged MSD (EATAMSD, black dashed line) on a log–log scale.

5.3 Physical Interpretation

The agreement between EAMSD and EATAMSD demonstrates that the system is **ergodic**. This means:

- Statistical properties obtained from a single long trajectory are equivalent to those obtained from many independent trajectories.
- The stochastic process is stationary and does not exhibit memory effects or aging.

Thus, the numerical simulation not only reproduces Brownian motion but also confirms its fundamental statistical property of ergodicity.

6 Scaling Analysis and Estimation of Diffusion Coefficient

6.1 Theoretical Background

For a diffusive process, the ensemble-averaged time-averaged mean squared displacement follows a power-law behavior with respect to lag time:

$$\langle \delta^2(\Delta) \rangle \sim \Delta^\alpha \tag{24}$$

where α is the scaling exponent. For normal Brownian motion, $\alpha = 1$, and the relation reduces to

$$\langle \delta^2(\Delta) \rangle = 2D\Delta \tag{25}$$

Taking logarithm on both sides,

$$\log\langle \delta^2(\Delta) \rangle = \alpha \log \Delta + \log C \tag{26}$$

where $C = 2D$. This transforms the power-law relation into a linear form, enabling estimation of parameters using linear regression.

6.2 Methodology

To determine the scaling exponent and diffusion coefficient, a log–log transformation of the ensemble-averaged TAMSD was performed:

$$x = \log_{10}(\Delta), \qquad y = \log_{10}(\langle \delta^2(\Delta) \rangle) \tag{27}$$

A linear fit of the form

$$y = mx + c \tag{28}$$

was applied, where

- m represents the scaling exponent α,
- c represents the intercept, related to the prefactor C.

The diffusion coefficient is then obtained as

$$D = \frac{10^c}{2} \tag{29}$$

6.3 Results

From the linear fitting procedure:

- Scaling exponent:

$$\alpha = 1.0079 \tag{30}$$

- Intercept:

$$c = 0.0182 \tag{31}$$

- Estimated diffusion coefficient:

$$D \approx 0.5214 \tag{32}$$

The extracted value of the scaling exponent $\alpha \approx 1.0079$ is very close to unity, confirming that the simulated motion follows **normal Brownian diffusion**. The slight deviation from exactly 1 arises due to finite sampling and numerical fluctuations.

The estimated diffusion coefficient $D \approx 0.5214$ is in excellent agreement with the input value $D = 0.5$ used in the simulation. This validates both the numerical implementation of the Langevin dynamics and the statistical analysis based on TAMSD.

Combined with the earlier results showing agreement between ensemble-averaged MSD and time-averaged MSD, this confirms that the system exhibits **ergodic and diffusive behavior**, consistent with theoretical expectations for Brownian motion.

7 Ergodicity and Ergodicity Breaking Analysis

7.1 Concept of Ergodicity

In stochastic systems, **ergodicity** refers to the equivalence between time averages and ensemble averages. A process is said to be ergodic if statistical quantities computed from a single long trajectory are identical to those obtained from an ensemble of many independent trajectories.

In the present study, this property was already examined through the agreement between the ensemble-averaged MSD (EAMSD) and the ensemble-averaged time-averaged MSD (EATAMSD). To further quantify this property, we analyze the **ergodicity breaking (EB) parameter**, which provides a direct measure of deviations between time and ensemble averages.

7.2 Definition of Ergodicity Breaking Parameter

The ergodicity breaking parameter is defined in terms of the fluctuations of TAMSD across different trajectories:

$$EB(\Delta) = \frac{\langle \delta^4(\Delta) \rangle}{\langle \delta^2(\Delta) \rangle^2} - 1 \tag{33}$$

where

- $\delta^2(\Delta)$ is the TAMSD for a given trajectory,
- $\langle \cdot \rangle$ denotes averaging over all trajectories.

This expression quantifies the spread of TAMSD values across the ensemble.

7.3 Theoretical Expectation

For standard Brownian motion, the ergodicity breaking parameter follows the scaling relation:

$$EB(\Delta) = \frac{4\Delta}{3T} \tag{34}$$

where T is the total measurement time. This implies that:

- $EB \to 0$ as $\Delta \ll T$,
- ergodicity is recovered in the long-time limit.

7.4 Results and Discussion

The figure shows the variation of the ergodicity breaking parameter as a function of lag time Δ, along with the theoretical prediction.

Key observations:

- The EB parameter increases with lag time Δ, consistent with theoretical expectations.

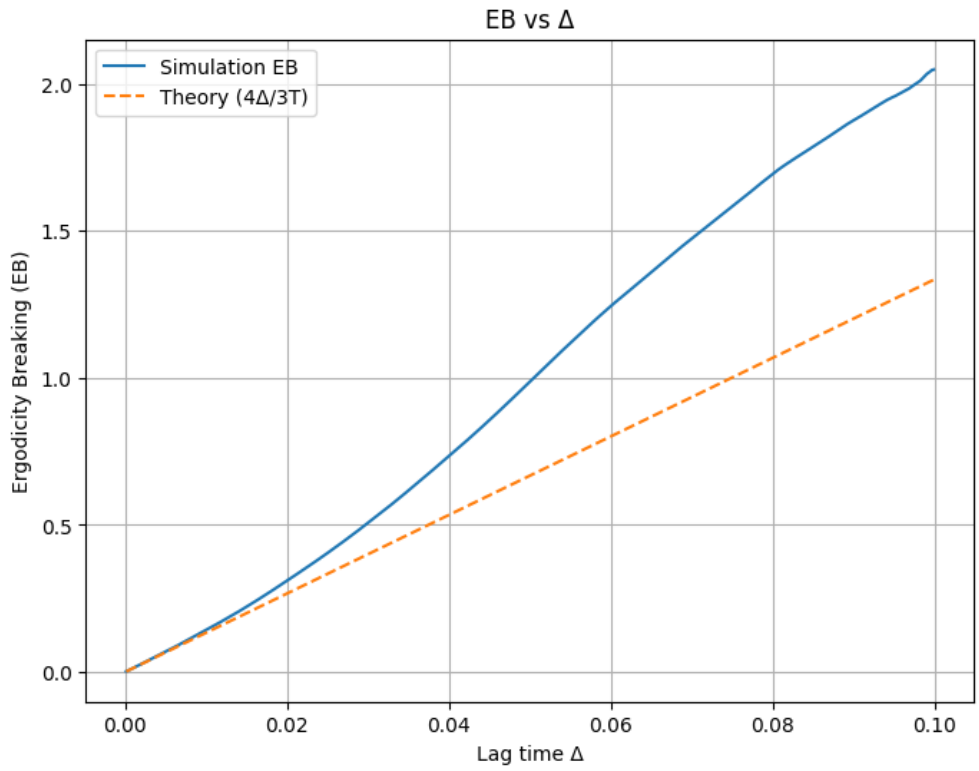

Figure 4: Variation of the ergodicity breaking parameter $EB(\Delta)$ as a function of lag time Δ, along with the theoretical prediction $EB(\Delta) = 4\Delta/3T$.

- At small lag times, the simulation closely follows the theoretical linear behavior.
- As Δ increases, the simulation curve deviates upward from the theoretical prediction.
- This deviation arises due to finite trajectory length and reduced statistical averaging at larger lag times.

7.5 Physical Interpretation

The non-zero values of EB indicate the presence of fluctuations in TAMSD across different trajectories. However, the overall agreement with the theoretical scaling confirms that the system approaches ergodic behavior in the limit of small Δ/T.

Combined with the earlier results:

- linear scaling of TAMSD,
- agreement between EAMSD and EATAMSD,

the EB analysis further supports that the simulated system exhibits the statistical properties of **Brownian motion with weak finite-time deviations**.

8 Distribution of Particle Positions and Gaussian Analysis

8.1 Theoretical Background

For a Brownian particle undergoing diffusion, the probability distribution of position at time t is governed by the diffusion equation. The solution of this equation for an initial condition $x(0) = 0$ is a Gaussian distribution:

$$P(x,t) = \frac{1}{\sqrt{4\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right) \tag{35}$$

This result implies that:

- The mean position remains zero:

$$\langle x(t) \rangle = 0 \tag{36}$$

- The variance grows linearly with time:

$$\langle x^2(t) \rangle = 2Dt \tag{37}$$

Thus, at any fixed time t, the distribution of particle positions across an ensemble is expected to follow a Gaussian form.

8.2 Methodology

To verify this theoretical prediction, the positions of all trajectories at the final time $t = T$ were extracted:

$$x_i(T), \qquad i = 1, 2, \ldots, N_{\text{traj}} \tag{38}$$

A histogram of these positions was constructed to approximate the probability density function. The distribution was then fitted using a Gaussian function:

$$P(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \tag{39}$$

where μ and σ^2 were obtained directly from the data. Additionally, the theoretical distribution $P(x,t)$ was plotted for comparison.

8.3 Results

From the fitting procedure:

- Mean:

$$\mu \approx 0.0043 \tag{40}$$

- Variance:

$$\sigma^2 \approx 0.0988 \tag{41}$$

- Theoretical prediction:

$$2Dt = 0.0999 \tag{42}$$

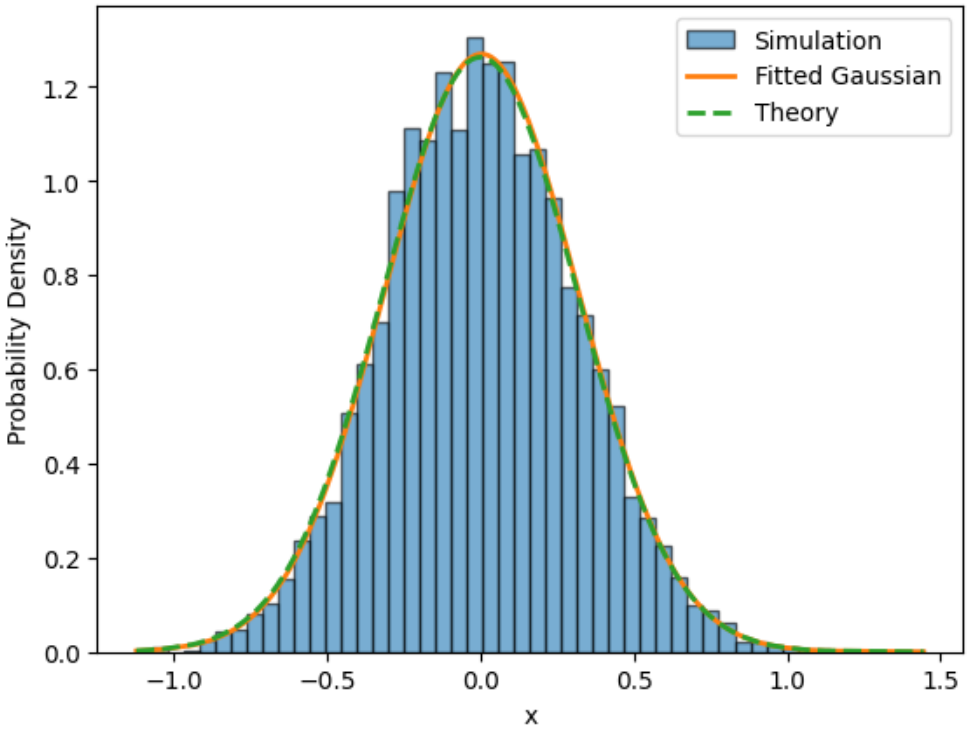

Figure 5: Histogram of particle positions at final time $t = T$ fitted with a Gaussian distribution, along with the theoretical prediction $P(x, t)$.

The histogram of particle positions shows a symmetric bell-shaped curve centered around zero, indicating no net drift in the system. The fitted Gaussian closely overlaps with both the simulation data and the theoretical prediction.

The extracted mean is very close to zero, confirming that the motion is unbiased. The variance obtained from the simulation matches remarkably well with the theoretical value $2Dt$, demonstrating the expected linear growth of fluctuations with time.

This result is fully consistent with the earlier findings:

- The TAMSD exhibited linear scaling with lag time.
- The ensemble-averaged MSD matched the time-averaged MSD.
- The scaling exponent was found to be approximately unity.

Together, these observations confirm that the simulated trajectories follow **classical Brownian motion**, and the statistical properties are accurately captured both in time and ensemble descriptions.

8.4 Physical Interpretation

The Gaussian distribution of positions arises from the cumulative effect of many independent random displacements. Each step contributes a small random increment, and by the central limit theorem, the overall distribution converges to a Gaussian form.

Thus, this analysis provides a direct verification of the fundamental statistical nature of diffusion and completes the characterization of the system from trajectory level to distribution level.

9 Time Evolution of Variance and Diffusion Validation

9.1 Theoretical Background

For a Brownian particle undergoing diffusion, the variance of position evolves linearly with time. Starting from the diffusion equation, the second moment is given by

$$\sigma^2(t) = \langle x^2(t) \rangle - \langle x(t) \rangle^2 \tag{43}$$

For an unbiased process $\langle x(t) \rangle = 0$, this reduces to

$$\sigma^2(t) = \langle x^2(t) \rangle \tag{44}$$

From diffusion theory, one obtains

$$\sigma^2(t) = 2Dt \tag{45}$$

which provides a direct relationship between variance and time.

9.2 Methodology

To verify this relation, the variance of particle positions was computed at each time step across the ensemble of trajectories:

$$\sigma^2(t_j) = \frac{1}{N_{\text{traj}}} \sum_{i=1}^{N_{\text{traj}}} \left[x_i(t_j) - \langle x(t_j) \rangle \right]^2 \tag{46}$$

where t_j denotes discrete time steps. This corresponds to calculating the spread of particle positions at each time across all trajectories. The resulting variance was then compared with the theoretical prediction $2Dt$.

9.3 Results

The figure shows the evolution of variance $\sigma^2(t)$ obtained from simulation along with the theoretical curve $2Dt$.

Key observations:

- The simulated variance increases linearly with time.
- The simulation curve closely follows the theoretical prediction over the entire time range.
- No significant deviation is observed, even at larger times.

9.4 Physical Interpretation

The linear growth of variance confirms that the system follows normal diffusive behavior. This result is consistent with all previous analyses:

- TAMSD showed linear scaling with lag time.

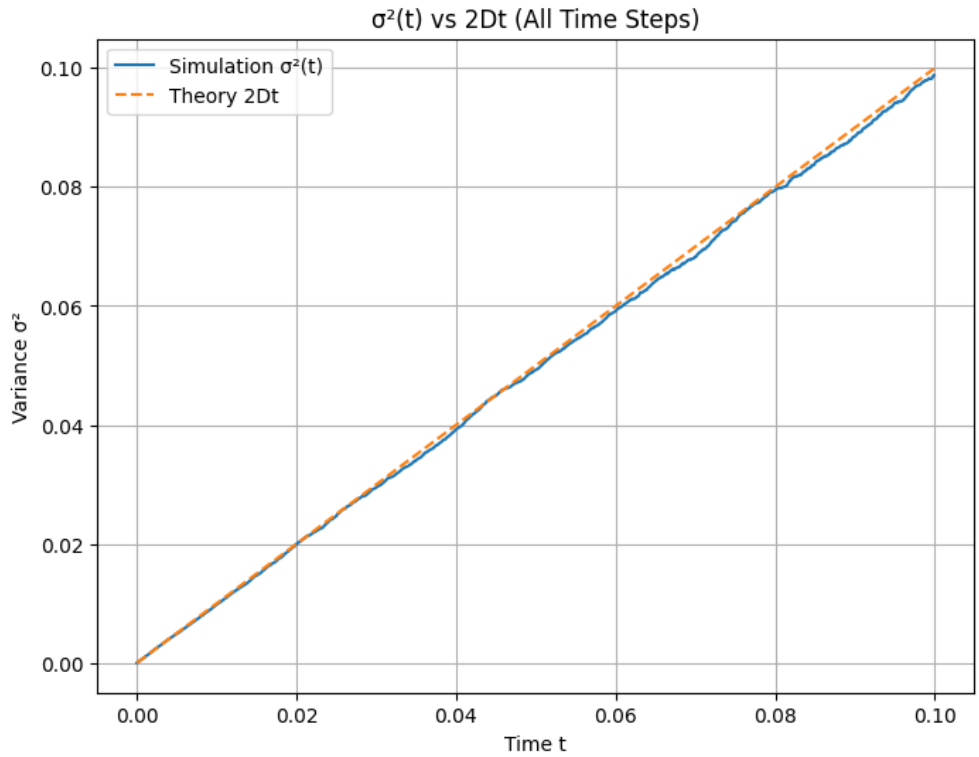

Figure 6: Time evolution of variance $\sigma^2(t)$ obtained from simulation (colored curve) compared with the theoretical prediction $2Dt$ (dashed line).

- EAMSD matched EATAMSD, confirming ergodicity.
- The position distribution followed a Gaussian form.

The present analysis provides a direct verification of the fundamental diffusion law $\sigma^2(t) = 2Dt$, independently from TAMSD and distribution-based methods.

Thus, the simulation accurately reproduces the expected statistical behavior of Brownian motion, and the diffusion coefficient is consistently validated across multiple independent approaches.

10 Time Evolution of Probability Distribution Function

10.1 Theoretical Background

For a diffusive process, the probability density function (PDF) of particle position evolves with time according to the diffusion equation. For an initial condition $x(0) = 0$, the solution is given by

$$P(x,t) = \frac{1}{\sqrt{4\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right) \tag{47}$$

This expression represents a Gaussian distribution whose width increases with time. The variance of the distribution is

$$\sigma^2(t) = 2Dt \tag{48}$$

indicating that the spread of particle positions grows linearly with time.

10.2 Methodology

To examine the time evolution of the distribution, particle positions were extracted at multiple time instances:

$$t = t_1, t_2, t_3, \ldots \tag{49}$$

For each selected time, a histogram of positions across all trajectories was constructed to approximate the PDF. A Gaussian fit of the form

$$P(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \tag{50}$$

was applied to the simulated data. In parallel, the theoretical distribution $P(x,t)$ was computed using the known diffusion coefficient.

10.3 Results

The figure shows the probability distributions at different times along with the corresponding fitted Gaussian and theoretical curves.

Key observations:

- At all time instances, the simulated distributions exhibit a symmetric Gaussian shape centered around zero.
- The fitted Gaussian curves closely match the histogram data, confirming that the distribution remains Gaussian at all times.
- The theoretical curves overlap with both the simulation and fitted distributions, indicating excellent agreement with diffusion theory.
- As time increases, the distributions broaden significantly, reflecting the increase in variance $\sigma^2(t) \propto t$.

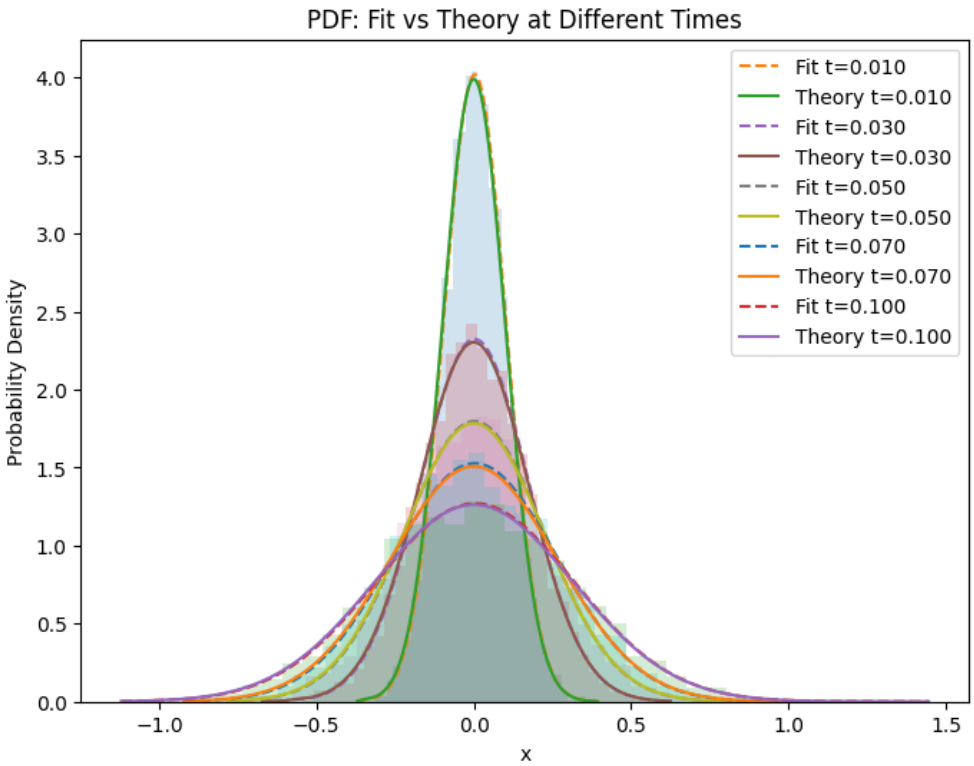

Figure 7: Probability distributions of particle positions at different time instances, along with the corresponding fitted Gaussian and theoretical curves $P(x, t)$.

- The peak height decreases with time, which is consistent with normalization of the probability density as the distribution spreads.

10.4 Physical Interpretation

The observed behavior demonstrates that the diffusion process preserves its Gaussian nature over time, while the width of the distribution increases due to cumulative random displacements.

This result is consistent with all previous analyses:

- The variance was shown to grow linearly with time.
- The MSD and TAMSD exhibited linear scaling.
- The scaling exponent was found to be unity.

Thus, the time evolution of the PDF provides a complete confirmation of the diffusion process, showing that the system follows classical Brownian motion not only in moments (MSD, variance) but also in the full probability distribution.

11 Displacement Distribution Analysis at Different Lag Times

11.1 Theoretical Background

For a Brownian particle, the displacement over a finite lag time Δ is defined as

$$\Delta x = x(t + \Delta) - x(t) \tag{51}$$

For a diffusive process, this displacement follows a Gaussian distribution with zero mean and variance proportional to the lag time:

$$P(\Delta x, \Delta) = \frac{1}{\sqrt{4\pi D\Delta}} \exp\left(-\frac{(\Delta x)^2}{4D\Delta}\right) \tag{52}$$

Thus, the variance of the displacement is

$$\langle (\Delta x)^2 \rangle = 2D\Delta \tag{53}$$

This implies that as the lag time increases, the distribution broadens while remaining Gaussian.

11.2 Methodology

To analyze this behavior, displacements were computed for multiple lag values:

$$\Delta = 2, 10, 15, 20, 50, 60, 70, 100 \tag{54}$$

For each lag:

- Displacements were calculated over all trajectories and all valid time indices.
- A histogram was constructed to estimate the probability density function.
- A Gaussian fit was performed using the simulated data.
- The theoretical distribution was computed using $\sigma^2 = 2D\Delta\, dt$.

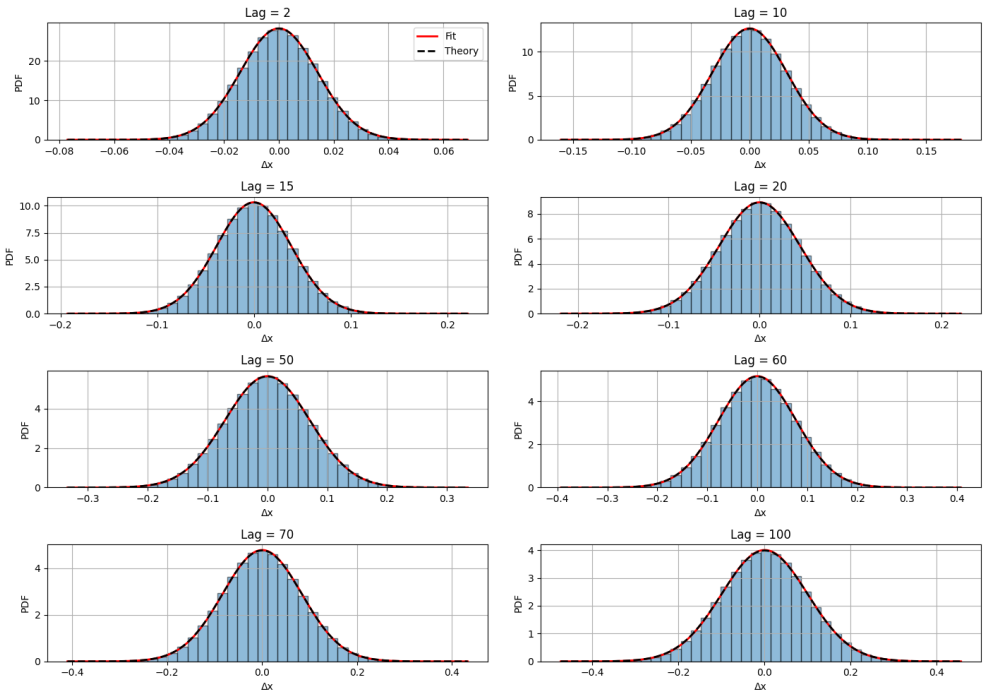

Figure 8: Displacement distributions for different lag values along with fitted Gaussian and theoretical curves.

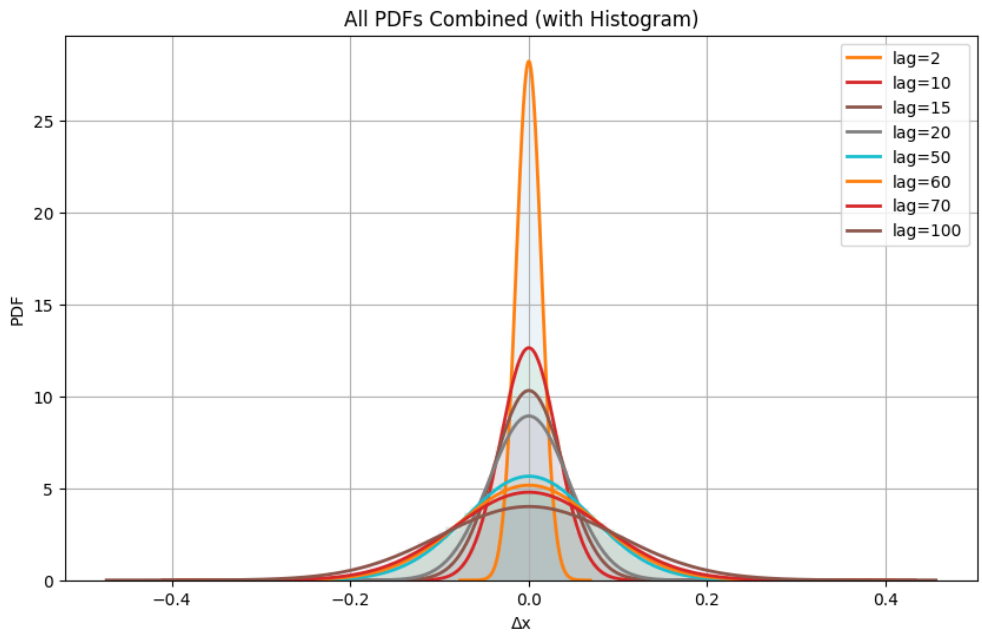

Figure 9: Combined plot of displacement distributions for all lag values, showing the progressive broadening with increasing lag time Δ.

11.3 Results and Discussion

The first set of plots shows the displacement distributions for different lag values along with fitted and theoretical curves.

Key observations:

- For all lag values, the displacement distributions exhibit a symmetric Gaussian shape centered at zero.
- The fitted Gaussian curves closely match the simulation histograms, confirming the validity of the Gaussian assumption.
- The theoretical curves overlap with the fitted distributions, indicating strong agreement with diffusion theory.
- As the lag increases, the distributions become broader and flatter, reflecting the increase in variance:

$$\sigma^2 \propto \Delta \tag{55}$$

The combined plot further highlights this behavior:

- Narrow, sharply peaked distributions are observed for small lag values.
- Wider distributions with lower peak height appear for larger lag values.
- All curves remain centered at zero, confirming the absence of drift.

11.4 Physical Interpretation

The increase in width of the distributions with lag time reflects the accumulation of random displacements over longer time intervals. Despite this increase, the Gaussian nature of the distribution is preserved, which is a direct consequence of the central limit theorem.

This result is consistent with previous findings:

- The variance was shown to scale linearly with time.
- The MSD and TAMSD followed linear scaling.
- The position distributions were Gaussian at all times.

Thus, the displacement-based analysis provides a complementary validation of diffusion, confirming that the process is Gaussian at all time scales and that the variance grows proportionally with lag time.

12 Results and Summary

In this study, the stochastic motion of a Brownian particle was simulated using a discrete Langevin model, and its statistical properties were analyzed through multiple complementary approaches.

The generated trajectories exhibited random behavior with no preferred direction, confirming the absence of drift. As time progressed, the spread of trajectories increased, indicating diffusive motion. This qualitative observation was quantitatively verified through the mean squared displacement (MSD), which showed a linear dependence on time:

$$\langle x^2(t) \rangle \propto t \tag{56}$$

The time-averaged mean squared displacement (TAMSD) was computed for individual trajectories and subsequently averaged over the ensemble. The TAMSD displayed linear scaling with lag time, consistent with normal diffusion. Furthermore, the excellent agreement between ensemble-averaged MSD (EAMSD) and ensemble-averaged TAMSD (EATAMSD) confirmed that the system is **ergodic**, meaning time and ensemble averages are equivalent.

A scaling analysis of TAMSD yielded a power-law exponent $\alpha \approx 1$, verifying that the system follows classical Brownian motion. The diffusion coefficient estimated from this analysis was found to be in close agreement with the input value used in the simulation, demonstrating the accuracy of the numerical approach.

The ergodicity breaking (EB) parameter was also evaluated to quantify fluctuations between trajectories. The EB parameter followed the expected linear trend with lag time and remained small for short times, further supporting the ergodic nature of the system, with minor deviations arising due to finite-time effects.

The probability distribution of particle positions at fixed times was found to be Gaussian, with mean approximately zero and variance growing linearly with time:

$$\sigma^2(t) = 2Dt \tag{57}$$

This was verified both through direct variance calculation and through fitting the distribution to a Gaussian function. The agreement between simulation and theoretical predictions confirms the validity of the diffusion model.

Additionally, the time evolution of the probability distribution demonstrated that the Gaussian shape is preserved at all times while the width increases with time. This behavior reflects the cumulative effect of random displacements.

The displacement distributions for different lag times were also analyzed. For all lag values, the distributions remained Gaussian and centered at zero, with variance increasing proportionally to the lag:

$$\langle (\Delta x)^2 \rangle \propto \Delta \tag{58}$$

This confirms that the process remains statistically consistent across different time scales.

Overall, all analyses — including trajectory behavior, MSD, TAMSD, scaling exponent, ergodicity measures, position distributions, variance evolution, and displacement distributions — consistently support the conclusion that the simulated system exhibits **classical Brownian motion with normal diffusion and ergodic behavior**.

13 Conclusion and Future Scope

This project successfully demonstrated the numerical simulation and statistical analysis of Brownian motion using a discrete Langevin dynamics model. Through multiple complementary approaches, the fundamental properties of classical Brownian diffusion were verified with high accuracy.

The key conclusions drawn from this study are as follows. The simulated trajectories exhibited purely random behavior with no preferred direction, confirming the absence of drift. The TAMSD showed linear scaling with lag time, and the estimated diffusion coefficient $D \approx 0.5214$ agreed closely with the input value $D = 0.5$. The scaling exponent $\alpha \approx 1$ confirmed normal diffusion. The agreement between EAMSD and EATAMSD established the ergodic nature of the system, and the ergodicity breaking parameter followed the theoretical prediction $EB(\Delta) = 4\Delta/3T$ at short lag times.

The probability distributions of particle positions were Gaussian at all times, with variance growing as $\sigma^2(t) = 2Dt$, consistent with diffusion theory. Displacement distributions at multiple lag times further confirmed the Gaussian and ergodic character of the process across all time scales.

13.1 Future Scope

While this project focused on one-dimensional Brownian motion, several interesting directions can be explored in the future:

- Extension to **two-dimensional** and **three-dimensional** Brownian motion to study anisotropic diffusion.
- Simulation of **anomalous diffusion** processes such as subdiffusion ($\alpha < 1$) and superdiffusion ($\alpha > 1$) using fractional Brownian motion or continuous-time random walk models [2].
- Study of **confined diffusion** in bounded domains, relevant to intracellular transport in biological systems [1].
- Application of **Bayesian model selection** techniques to distinguish between different diffusion models from trajectory data.
- Investigation of **non-Gaussian diffusion** and ergodicity breaking in heterogeneous media.

Code Availability

All simulation codes and analysis scripts associated with this project are publicly available on GitHub at the following repository:

`https://github.com/vivek-meena/Interpolation-of-fractional-Brownian-motion`

The repository contains Python scripts and a Jupyter Notebook (`1ALL_CODE.ipynb`) covering all simulations and analyses presented in this report, including trajectory generation, TAMSD computation, MSD analysis, Gaussian fitting, ergodicity breaking parameter evaluation, and diffusion coefficient estimation. All codes are implemented using `NumPy` and `Matplotlib` and can be executed locally with standard Python dependencies.

References

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[2] I. M. Jánosi, A. Padash, J. A. C. Gallas, and H. Kantz, "Passive tracer advection in the equatorial Pacific region: statistics, correlations and a model of fractional Brownian motion," *Ocean Sci.*, vol. 18, pp. 307–320, 2022. https://doi.org/10.5194/os-18-307-2022

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